Dallas County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 9, 2026 · 32.9535, -96.7700

SIS sample draft — full tier

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) presents the anticipated traffic impacts of the proposed Dallas County Sample — Shopping Center / Retail located within Dallas-Fort Worth-Arlington MSA, Texas. The study evaluates 4 intersections within a 0.50-mile study radius. The report follows the outline in TxDOT Traffic and Safety Analysis Procedures Manual (TSP) Chapter 16 Appendix Q, with capacity analysis per the Highway Capacity Manual (HCM) latest edition and trip generation per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716). The development generates a determining peak-hour trip count of 680 vehicles, classifying it as Category 2 (500–1,000 peak-hour trips) under TSP §16.2.1 Table 16-1.

Reviewing authority: City of Dallas.

Dallas Department of Transportation — Traffic Engineering, per Dallas Development Code §51A-4.803 (Site Plan Review) and Connect Dallas (Strategic Mobility Plan, adopted Apr 28, 2021).

Findings:

• 0 intersections project to drop one or more LOS grade under the Background-plus-site scenario.

• 2 intersections operate at LOS E or F under the Background-plus-site scenario and are flagged for mitigation under TSP §16.4.3.

4	0	2	0.7s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

1.0 INTRODUCTION

Per TSP §16.1, this Traffic Impact Analysis determines whether the transportation infrastructure surrounding the project can accommodate the traffic demand the proposed development will introduce. It is layered with the City of Dallas TIA standard where applicable. Dallas Department of Transportation — Traffic Engineering, per Dallas Development Code §51A-4.803 (Site Plan Review) and Connect Dallas (Strategic Mobility Plan, adopted Apr 28, 2021).

TSP §16.1 also flags that this chapter does not provide final TIA recommendation thresholds — the TxDOT District and the host municipality may request analysis beyond this scope, and the project manager determines final methodology. Refer to the TxDOT Access Management Manual Chapter 3 for the TIA-request criteria that apply on the state system.

Preliminary Scoping (TSP §16.2.2)

Per TSP §16.2.2, the specific project scope should be confirmed with the local TxDOT District before the TIA begins, via an in-person meeting or alternate correspondence. Items to confirm:

• Number of access points to TxDOT roadways
• Acceptable LOS thresholds
• Selected study years
• Anticipated influence area
• Intersections and roadways to be analyzed
• Scenarios to analyze
• Data collection method
• Project schedule and buildout year
• Data source
• Use of TDM outputs, growth factors, etc.
• Other major projects in the area

Required submission: No fixed required-elements list. Submittal carries the TxDOT-equivalent engineering tables/figures plus alignment with Connect Dallas and any PD-overlay traffic conditions. A TIS Waiver form is required when the threshold is not triggered.

Host-jurisdiction TIA threshold: Less than 1,000 trips per day → no Traffic Impact Study or Waiver required (Paving/Drainage TIS Waiver form). Greater than 1,000 trips per day → either a Traffic Impact Study or a TIS Waiver is required; waivers are considered per-case by the Director of the Department of Development Services.

Dallas TIA standards are not consolidated into a single dated manual — review is partly discretionary under §51A-4.803 site plan review. This report aligns with the engineering tables and figures expected by Dallas DOT Traffic Engineering plus the multimodal context of Connect Dallas.

2.0 PROJECT DESCRIPTION

2.1 Site Plan

Project name	Dallas County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	32.9535°, -96.7700°
Host jurisdiction	City of Dallas

2.2 Area of Influence

The anticipated area of influence is the 0.50-mile radius surrounding the site, defined per TSP §16.3.1 to include the roads and intersections within the project and the area impacted by project-generated trips. Final area of influence is confirmed with the District during preliminary scoping.

2.3 Phasing and Timing

Single-phase buildout year: 2027. Future analysis horizon for this TIA category: Buildout year, each phase completion year, and five years past buildout (TSP §16.2.1 Category 2).

Phased development scenarios are not modeled in this screening analysis. Each phase completion year listed above should be analyzed separately in the formal submittal — phase boundaries and completion dates must come from the applicant's construction schedule.

3.0 STUDY AREA CONDITIONS

3.1 Existing and Anticipated Land Use

Existing land use and the host-jurisdiction Future Land Use Plan for parcels within the area of influence should be compiled from the host city's adopted Comprehensive Plan and any applicable overlay districts. This screening report does not generate that inventory [placeholder — requires GIS pull against host-jurisdiction parcel layer].

3.2 Existing and Future Roadway System

Roadway functional class, lane count, posted speed, and surface for state-system routes (IH / US / SH / FM / RM / BU / BS / SL / SS) are referenced against TxDOT Roadway Inventory (RHiNo) and the TxDOT Statewide Planning Map. Future roadway system context is taken from the TxDOT Unified Transportation Program (UTP) and the regional MPO Metropolitan Transportation Plan (MTP) where in-area projects are programmed.

4.0 EXISTING OPERATIONS

4.1 Roadway Conditions and Traffic Controls

Geometry, signal timing, and traffic-control inventory for the study network are derived from RHiNo plus host-jurisdiction signal-timing records. For the formal submittal, supplementary field inspection within 12 months of submittal is recommended.

4.2 Alternate Modes

Pedestrian, bicycle, and transit facility inventories within the area of influence should be confirmed against host-jurisdiction GIS and regional transit-operator maps. Per TSP §16.3.3, multimodal reduction is applied to trip generation in areas where alternate transit is readily available — this screening report does not auto-apply multimodal reduction.

4.3 Traffic Volumes

Existing AADT for state-system segments is taken from the TxDOT Open Data Portal AADT layer (annual refresh). Peak-hour turning-movement counts at study intersections should be collected midweek (Tue/Wed/Thu), school-in-session, within 12 months of submittal — per Houston IDM §15.06.01.A counts must be within 12 months in high-growth areas and within 24 months elsewhere; Austin TDS no longer accepts pre-COVID counts by default.

4.4 Level of Service

Affected intersection	Distance (mi)	Existing LOS	Existing delay (s)
North Coit Road & West Belt Line Road	0.17	F	120.0
North Coit Road & Roundrock Road	0.23	B	14.1
West Belt Line Road & Dublin Drive	0.38	B	14.1
South Coit Road & Dumont Drive	0.49	F	120.0

4.5 Safety

Crash history for the most recent five years on study segments and intersections should be pulled from the TxDOT Crash Records Information System (CRIS Public Query). This screening report does not auto-generate the crash summary [placeholder — requires CRIS pull].

5.0 PROJECTED TRAFFIC

5.1 Site Generated Traffic

Trip generation is calculated per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at a proposed size of 85 1,000 sqft GLA. Per TSP §16.3.3, the fitted curve equation is preferred when the data plot contains at least 20 data points or has an R^2 of at least 0.75; otherwise average rates apply. Daily, AM peak, and PM peak hour trips are reported below.

Period	Entering trips	Exiting trips
Daily	3,400	3,400
AM peak hour	166	106
PM peak hour	326	354

Internal capture (per TSP §16.3.3: trips between land uses within the same development that do not touch the off-site street system) and pass-by trips (already traveling on the adjacent roadway network and entering the proposed development) are accounted for below.

Pass-by capture applied	30%
Internal capture applied	0%

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

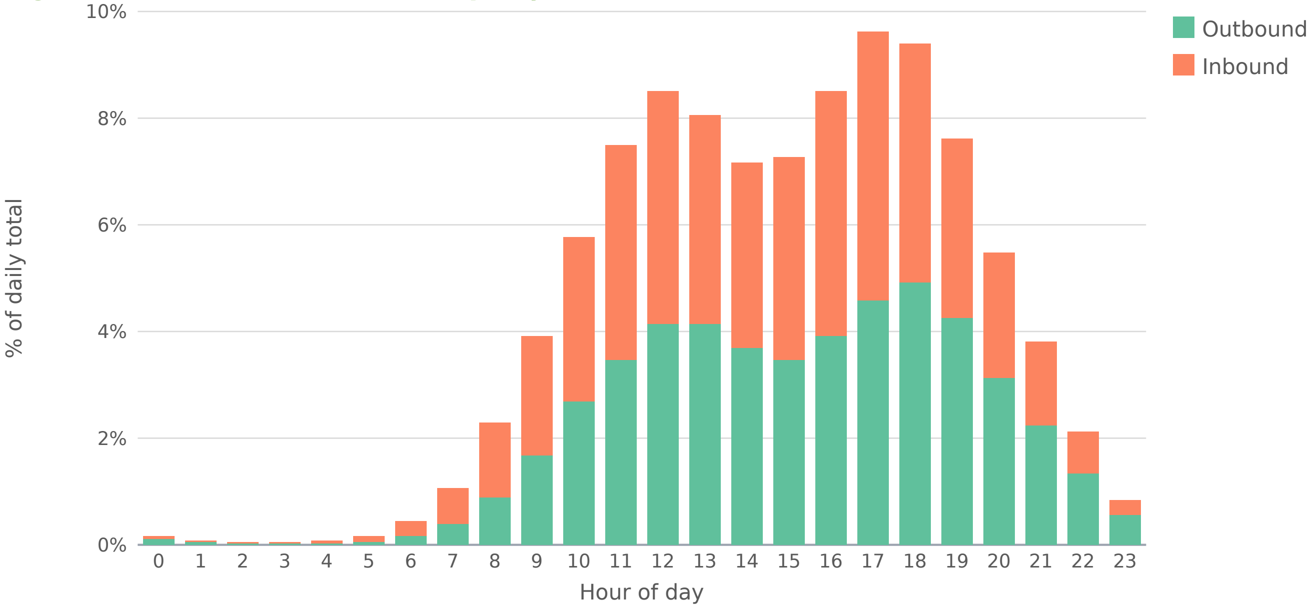
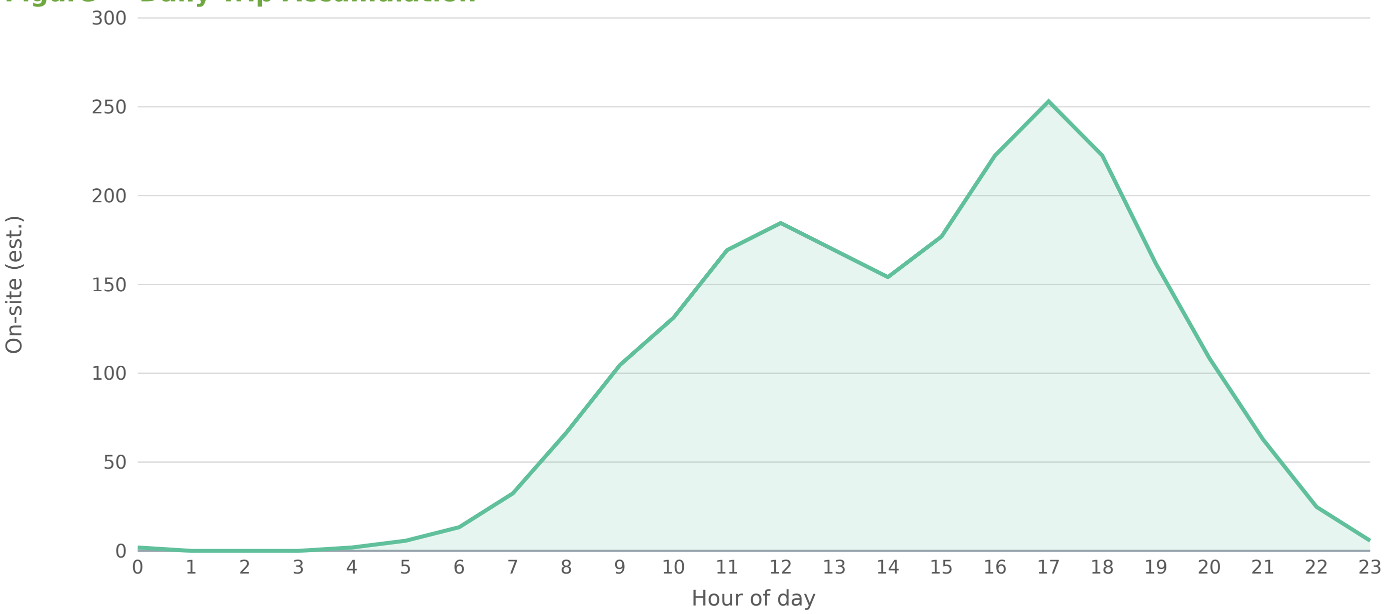


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

5.2 Trip Distribution

Per TSP §16.4, project distribution is assigned using engineering judgment, informed by the surrounding roadway network geometry and proximity to the project access points. If only one project driveway is proposed, all trips enter and exit through that driveway. Final distribution percentages are confirmed with the District during preliminary scoping.

5.3 Trip Assignment

Site-generated external trips are assigned to the study network using inverse-distance weighting from the project site to each signalized intersection, normalized so the period total matches the external-trip count from §5.1.

5.4 Non-Site Traffic

Background traffic is grown at 1.5% per year over 1 year. Per TSP §16.4.2, the prescribed method is to average at least the last five years of historical AADT data for the segment analyzed to derive an average annual growth rate; the value applied here is a screening default and should be re-calibrated to the affected segments' five-year AADT trend before formal submittal. Background growth data is also commonly sourced from the host city or regional MPO travel-demand model (H-GAC, NCTCOG, CAMPO, or AAMPO depending on the MSA).

Other major projects in the area (committed developments) must be coordinated with the District and governing municipality per TSP §16.3.3 and added on top of the AADT-derived background growth. This screening analysis does not auto-pull committed-development trips.

5.5 Total Traffic

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	227	139	88
PM Peak Hour	680	204	0	430	206	224
Saturday Midday	748	56	0	625	312	313
Daily Total	6,800	510	0	5,678	2,839	2,839

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross − pass-by − internal-capture) × auto-mode share (90.1% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

5.6 Trip Distribution — Gravity Model

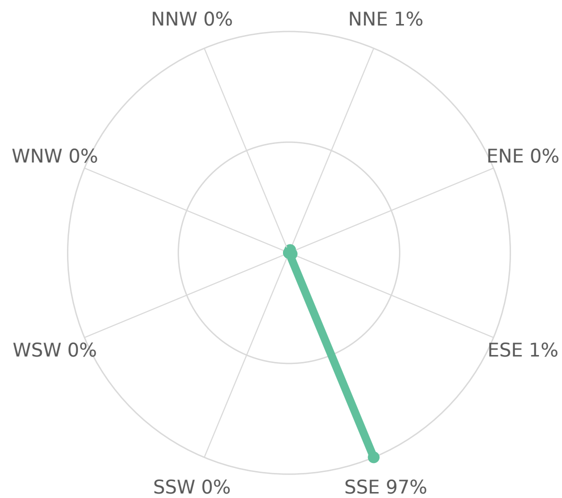
The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	1%
Southeast (ESE + SSE)	99%
Southwest (SSW + WSW)	0%
Northwest (WNW + NNW)	0%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
North Coit Road & West Belt Line Road	SSE	0.17	3,747	0.5	50.86%
South Coit Road & Dumont Drive	SSE	0.49	3,872	0.5	46.55%
North Coit Road & Roundrock Road	NNE	0.23	100	0.0	1.33%
West Belt Line Road & Dublin Drive	ESE	0.38	100	0.0	1.26%

4 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

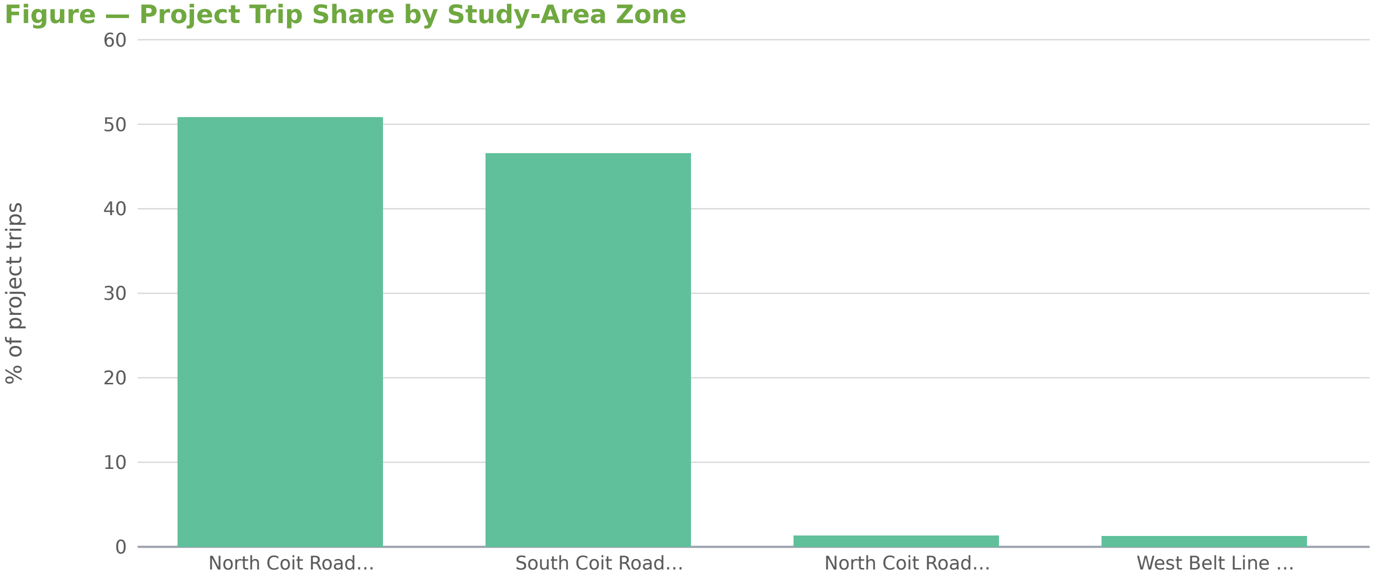
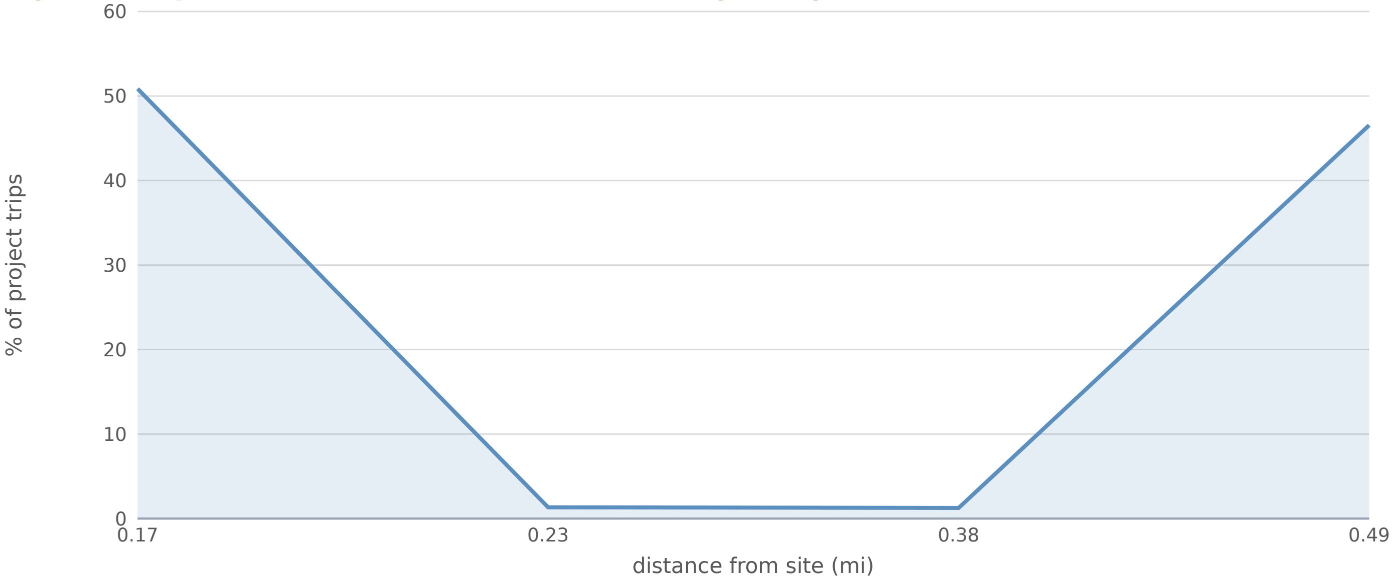


Figure — Trip Share vs. Distance from Site (Gravity Decay)



6.0 TRAFFIC AND IMPROVEMENT ANALYSIS

6.1 Site Access

Where the project fronts a state-system roadway (IH / US / SH / FM / RM / BU / BS / SL / SS), site access is reviewed through the TxDOT Driveway Access Permit (DAP) process under the Access Management Manual Chapter 3 §3. Driveway spacing, geometry, and auxiliary lanes (right-turn deceleration, left-turn) are checked against ACM Chapter 2 §3 (Table 2-2 spacing) and the Roadway Design Manual Chapter 16.

6.2 Auxiliary Lane and Sight Distance Analysis

Auxiliary lane warrants (right-turn deceleration, left-turn) per ACM Chapter 2 §4 and Roadway Design Manual Chapter 16 should be checked against the project's access geometry. Intersection sight distance per RDW Ch. 2 should be verified at every proposed driveway. This screening report does not perform per-driveway sight-distance or auxiliary-lane warrant calculations [placeholder — requires final driveway geometry].

6.3 Capacity and Level of Service Analysis

Per TSP §16.4.1 and §16.4.4, capacity analysis follows the latest HCM methodology using one of the District-accepted tools (Synchro, HCS, Vissim, or Vistro). For signalized intersections, each approach and the overall intersection are analyzed. Two future scenarios are compared at each affected intersection: Background (grown traffic without the proposed project) and Background-plus-site (grown traffic plus the project's external trips at the assigned distribution). An Existing scenario is included for context. Host-jurisdiction LOS standard: Transitional — Connect Dallas (Apr 2021) is moving Dallas from LOS toward VMT. Practice currently uses LOS D suburban / LOS E in the CBD.

Intersection	Existing LOS	Background LOS	Bgd+Site LOS	Δ delay (s)	Q95 (ft)
North Coit Road & West Belt Line Road	F	F	F	0.0	1,352
North Coit Road & Roundrock Road	B	B	B	0.7	59
West Belt Line Road & Dublin Drive	B	B	B	0.3	36
South Coit Road & Dumont Drive	F	F	F	0.0	1,292

6.4 Mitigation

Per TSP §16.4.3, any operational deficiencies found in the future analysis are considered for mitigation. The threshold for acceptable operations (LOS, queue length, travel time, and other MOEs) is agreed upon with the District during preliminary scoping rather than being set as a single statewide standard. Typical mitigation measures listed in TSP §16.4.3 include: right-turn deceleration lanes, left-turn lanes, median modifications, traffic signal modification and installation, road widening, revised striping, turning lane restrictions, and alternative intersections / interchanges. The developer is responsible for implementing the agreed-upon mitigation measures.

Screening Mitigation Recommendations

North Coit Road & West Belt Line Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

South Coit Road & Dumont Drive [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

6.5 Driveway Operational Analysis

Per-driveway operational analysis (full-movement vs. left-in/left-out configuration, throat depth, on-site queue spillback) depends on the final site plan and is not included in this screening-level TIA [placeholder].

7.0 SAFETY ANALYSIS

Per the Appendix Q outline, Safety Analysis is a standalone section separate from §4.5 Existing Operations — Safety. It includes the project's effect on study-area crash trends, sight-distance impacts at proposed access, and any HSIP-identified crash clusters within the area of influence. This screening report does not auto-generate the safety analysis [placeholder — requires CRIS pull + sight-distance verification].

8.0 CONCLUSIONS

Based on the screening analysis above, the project is classified as Category 2 (500–1,000 peak-hour trips) per TSP §16.2.1. Of the 4 affected intersections analyzed, 0 drop one or more LOS grade and 2 operate at LOS E or F under the Background-plus-site scenario. The horizon analyzed in this screening is the buildout year only; the full submittal must cover the years required by the project's TIA category per TSP §16.2.1 Table 16-1.

9.0 RECOMMENDATIONS

Submit this TIA to City of Dallas, with parallel TxDOT-District coordination where any state-system route is in frontage. Validate the screening results against current-edition manuals, updated turning-movement counts within the most recent 12 months (Houston IDM §15.06.01.A: 12 months in high-growth areas, 24 months elsewhere), and the preliminary-scoping outcome with the District. The report must be sealed by a Texas-licensed Professional Engineer per 22 TAC §137.33 (Sealing Procedures), promulgated under The Texas Engineering Practice Act (Tex. Occ. Code Ch. 1001), with the seal on the cover and on every sealed sheet.

10.0 APPENDICES

The formal submittal appendices, per the Appendix Q outline, should include: scoping correspondence with the TxDOT District; site plan figures; TMC count sheets with date, weather, observer; Trip-generation worksheets with rate/equation selection rationale; HCS / Synchro / Vissim / Vistro output; signal warrant analyses citing the TMUTCD (2025 edition, effective Jan 18, 2026); auxiliary-lane and sight-distance worksheets; CRIS crash data summary; and the host-jurisdiction's submission forms (Houston Access Management Data Summary Form + CPC 101 Form / Austin TIA Determination + Scope / Dallas TIS Waiver / Fort Worth TIA Worksheet / San Antonio TIA Threshold Worksheet + Rough Proportionality calculation).

Programmed Projects (informational)

Review of programmed transportation projects within the area of influence should consult the TxDOT Unified Transportation Program (UTP 2026, adopted Aug 2025), the federally-required Statewide Transportation Improvement Program (STIP), and North Central Texas Council of Governments (NCTCOG) TIP and Mobility 2045. This screening analysis does not automatically integrate programmed-projects data; manual review is recommended for any submittal.

FINDINGS

• Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).

• Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).

• Auto-mode share of 90% applied to external trips for Dallas-Fort Worth-Arlington MSA; 10% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.

• Existing volumes were grown by 1.50%/yr over 1 year (×1.01) to the opening-year horizon.

• 4 signalized intersections fall within the study area; 0 are projected to drop at least one LOS grade after build-out.

• 2 intersections are projected to operate at LOS E or F under the build condition and require formal mitigation per Dallas Department of Transportation / Fort Worth Transportation & Public Works TIS guidance.

• Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a ±0.5%/yr background growth band around the applied value.

METHODOLOGY NOTES

• Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.

• Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.

• Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.

• Weather adjustment follows HCM 6th-Edition Ch. 11 (rain/snow capacity reduction): clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.

• Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: a production-constrained gravity model $T_j = P \cdot (A_j \cdot F_j) / \Sigma(A_x \cdot F_x)$ allocates trips to surrounding signals, where attractiveness A_j is the signal's through-volume and the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work coefficients a=28507, b=-0.02, c=-0.123) on the travel time t to each signal. Step 3 Mode Choice: a binary logit $P(auto)=1/(1+e^{-(ASC-\lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.

• After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.

• Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.

• Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.

• Intersection-level control delay uses the HCM signalized-intersection model d = d1 + d2 (Webster uniform delay + Akçelik/HCM incremental-delay term) with a 90s cycle, g/C = 0.45, 1,800 vphpl saturation flow (×

weather factor), 15-minute peak analysis period (T = 0.25 hr) and pretimed-signal incremental-delay factor k = 0.5. Reported control delay is capped at 120 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

• Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal allocation perturbed ±15% from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (HCM Eq. 19-50, $Q95 \approx Q1 \times 1.65 \times 25$ ft/veh) are reported.

• Level of Service is assigned from HCM 6th-Edition signalized-intersection control-delay thresholds (Exhibit 19-8): A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s.

• Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a ±0.5%/yr growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

• Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

• Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as (turbo lanes / approach lanes) × (1 − g/C) / (g/C), with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%…+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: a=28507, b=-0.02, c=-0.123) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
North Coit Road & West Belt Line Road	0.17	0.4	179	43.3%
North Coit Road & Roundrock Road	0.23	0.6	133	32.2%
West Belt Line Road & Dublin Drive	0.38	0.9	63	15.3%
South Coit Road & Dumont Drive	0.49	1.2	38	9.2%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 90% applied via a binary logit $P(auto) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 10% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 2.21. Per-signal v/c, delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

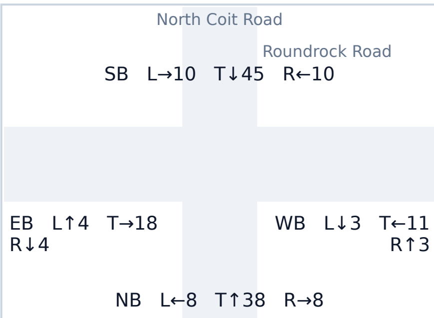
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 North Coit Road & Roundrock Road

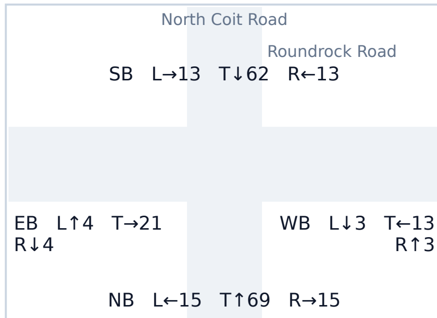
Signal ID	dallas-fort-worth-4027
Location	NE Dallas-Fort Worth-Arlington · 0.23 mi from site
Added PM peak trips	133
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.8 s/veh · v/c 0.13
Δ control delay	+0.7 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

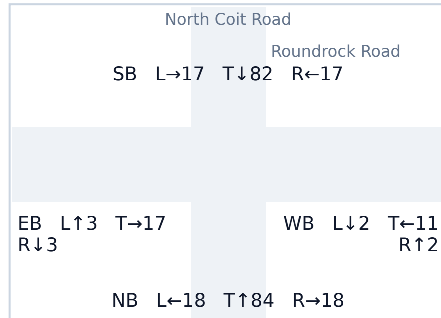
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	29	69	99	0.04 → 0.12	13.9 → 14.7	B → B	59
SB	24	64	88	0.03 → 0.11	13.9 → 14.6	B → B	53
EB	29	0	29	0.04 → 0.04	13.9 → 13.9	B → B	17
WB	19	0	19	0.02 → 0.02	13.8 → 13.8	B → B	11

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	69	52%
SB Through	64	48%

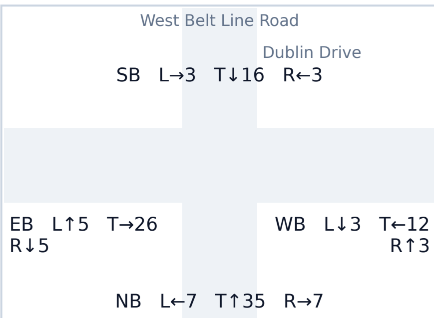
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 West Belt Line Road & Dublin Drive

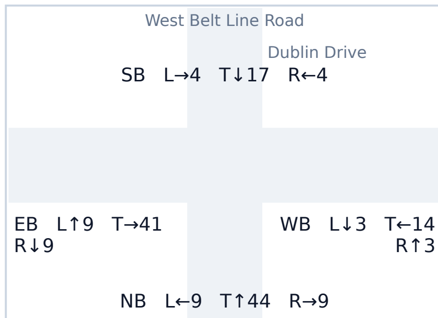
Signal ID	dallas-fort-worth-2678
Location	NE Dallas-Fort Worth-Arlington · 0.38 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.4 s/veh · v/c 0.09
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

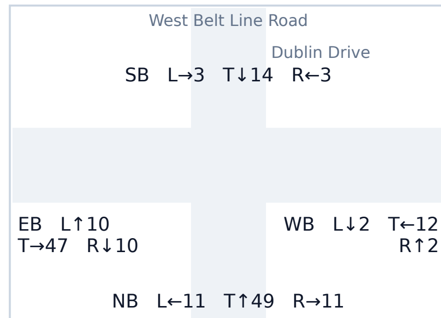
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	32	30	62	0.04 → 0.08	13.9 → 14.3	B → B	36
SB	25	0	25	0.03 → 0.03	13.9 → 13.9	B → B	14
EB	26	33	59	0.03 → 0.07	13.9 → 14.2	B → B	34
WB	20	0	20	0.02 → 0.02	13.8 → 13.8	B → B	12

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Right	32	51%
NB Left	30	48%
EB Through	1	2%

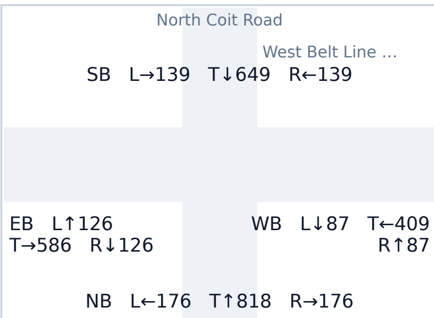
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 North Coit Road & West Belt Line Road

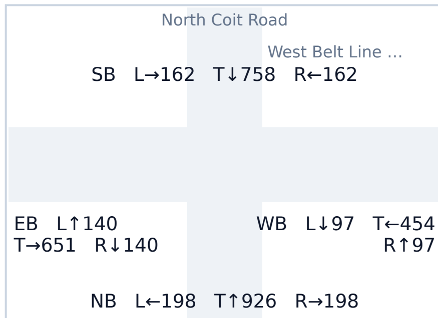
Signal ID	dallas-fort-worth-12880
Location	NE Dallas-Fort Worth-Arlington · 0.17 mi from site
Added PM peak trips	179
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 2.11
Build (PM)	LOS F · 120.0 s/veh · v/c 2.21
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

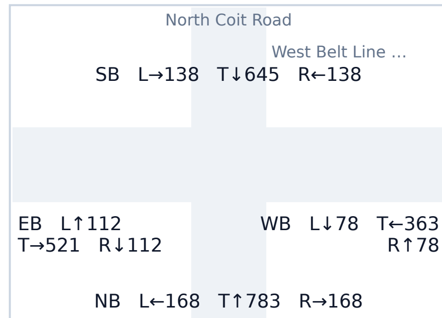
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,237	85	1,322	1.53 → 1.63	120.0 → 120.0	F → F	1,352
SB	989	93	1,082	1.22 → 1.34	120.0 → 120.0	F → F	1,107
EB	931	0	931	1.15 → 1.15	105.7 → 105.7	F → F	952
WB	647	1	648	0.80 → 0.80	29.4 → 29.4	C → C	574

Affected movements (PM peak project trips)

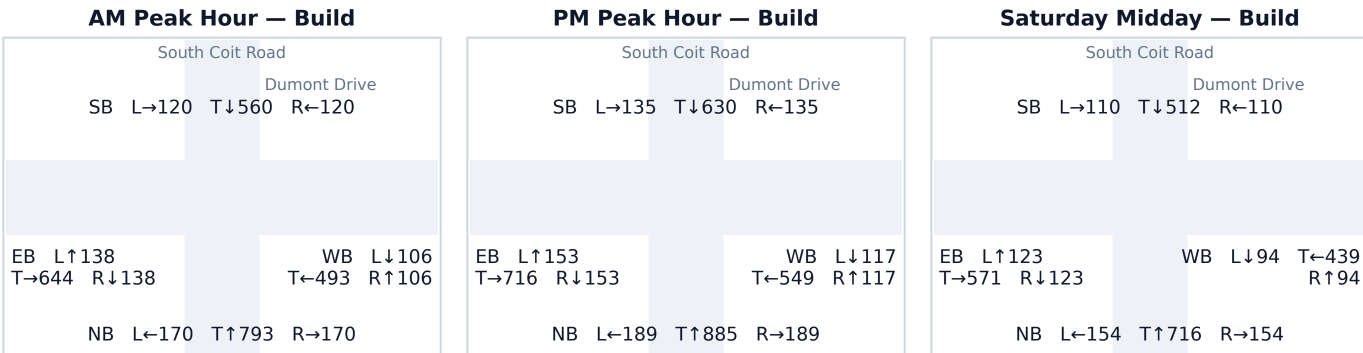
Movement	Project trips	% of project trips
SB Through	92	51%
NB Through	85	47%
SB Left	1	1%
WB Right	1	1%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 South Coit Road & Dumont Drive

Signal ID	dallas-fort-worth-761
Location	NE Dallas-Fort Worth-Arlington · 0.49 mi from site
Added PM peak trips	38
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 2.18
Build (PM)	LOS F · 120.0 s/veh · v/c 2.20
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,245	18	1,263	1.54 → 1.56	120.0 → 120.0	F → F	1,292
SB	880	20	900	1.09 → 1.11	82.3 → 91.1	F → F	920
EB	1,022	0	1,022	1.26 → 1.26	120.0 → 120.0	F → F	1,045
WB	783	0	783	0.97 → 0.97	48.6 → 48.6	D → D	786

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	20	53%
NB Through	18	47%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.